Eleventh Taiwan Olympiad in Linguistics

Round 1 Solutions

Problem 1 Eastern Arrernte

The word order is SOV. Ergative: **-le**. Absolutive: **∅**.

Tenses of verbs are as below:

Past: *STEM*-**ke**

Present: *STEM*-**me**

Future: *STEM*-**tyenhe**

Answer

(a)

9. *The child looks at the wedge-tailed eagle.*
10. *The emu came.*

(4 pts) [2 pts each.]

(b)

11. **arlewatyerrele marle atweme**
12. **intelyapelyape alkereke-iretyenhe**
13. **marlele akngwelye akaltyele-antetyenhe**

(8 pts) [2pts for question 12. 3pts each for question 11 and 13.]

(c)

14. **artwe akarelhetyenhe**
15. **intelyapelyapele apere untheke**
16. **artewele irretye arlkweme**

(8 pts) [2pts for question 14. 3pts each for question 15 and 16.]

Marking Criteria

For question 11 to 16, 1 pt for each argument. 0.5 pts are deducted for each spelling mistake that does not affect the meaning (i.e., irrelated to noun case suffix or verb tense suffix).

Reference

Green, Jenny. (2005). *A learner's guide to Eastern and Central Arrernte.* Alice Springs: IAD Press.

Problem 2 Thai

Answer

(a)

1.	**kæːŋ ka riː**	*Thai curry* | *curry*	F.	*yellow curry*
2.	**kæːŋ khiaw waːn**	*Thai curry* | *green* | *sweet*	G.	*green curry*
3.	**kæːŋ som**	*Thai curry* | *sour*	K.	*sour curry*
4.	**nam phrik**	*water* | *chili*	J.	*chili sauce*
5.	**nam waːn**	*water* | *sweet*	H.	*sweetened beverage*
6.	**siː khaːw**	*color* | *white*	D.	*white color*
7.	**siː som**	*color* | *citrus*	B.	*orange color*
8.	**som**		C.	*citrus*
9.	**som khiaw waːn**	*citrus* | *green* | *sweet*	I.	*green tangerine*
10.	**thuaː dæːŋ**	*bean* | *red*	A.	*red bean*
11.	**wai chɔːk koː læːt**		E.	*white chocolate*

(10 pts) [1 pt is deducted for each incorrect answer.]

(b)

12. *sweet*
13. *red chili* or *red pepper*

(2 pts) [1 pt each.]

(c)

14.	**thuaː khiaw**	*bean* | *green*
15.	**nam som**	*water* | *citrus*
16.	**siː nam**	*color* | *water*
17.	**chɔːk koː læːt khaːw**	*chocolate* | *white*

(8 pts) [2 pts each. For question 17, 1.5 pts are deducted for incorrect word order. 0.5 pts are deducted for each misspelling that does not affect the meaning.]

Reference

thai-language.com, http://www.thai-language.com/

Problem 3 Portuguese

The stress is on the syllable marked with ˆ or ´. If neither of these marks is present, the stress is on the penultimate syllable.

The pronunciations of **e**, **t** and **d** are as follows.

Portuguese orthography	**é**	**ê**	stressed **e**	unstressed **e**			**t**/**d**
				word initial	elsewhere	word final	before /**i**/
EP pronunciation	**ɛ**	**e**	**ɛ**	**i**	**ɨ**		**t**/**d**
BP pronunciation	**ɛ**	**e**	**ɛ**	**e**		**i**	**ʧ**/**ʤ**

Answer

(a)

1. **e'zo.fa.gu**
2. **ʤi.plo'ma.tɐ**
3. **fo.to.ɡɾa'fi.ɐ**
4. **'ɔ.ʧi.mu**

(4 pts) [1 pt each. 0 pts if multiple stresses are given in each question.]

(b)

5. **bɨ'bɐ̲**
6. **bɐ̲'bɐ̲**
7. **i̲.nɨ̲ɾ'ʒɛ̲.t̲i.ku**
8. **ɐ̲.nɐ̲ʁ'ʒɛ̲.ʧ̲i.ku**
9. **t̲ɐ'vɛ̲ɾ.nɐ**
10. **t̲a'vɛ̲ʁ.nɐ**
11. **vɨ̲ɾ'd̲a.d̲ɨ̲**
12. **vɐ̲ʁ'd̲a.ʤ̲i̲**

(16 pts) [1 pt for each phone.]

Reference

Wiktionary, https://www.wiktionary.org/

Problem 4 Karimata Miyako

The word order is SOV. Nominative: **-nu**. Accusative: **-u**.

The accusative case marker changes with the end of the noun stem as follows:

(1) **-**C + **-u** → **-**CC**u**

(2) **-**VV + **-u** → **-**VV**yu**

(3) **-**CV + **-u** → **-**CV**u**

(4) **-ɨ**: here **ɨ** reflects the consonant/vowel duality, and the changes are as follows:

If it is **-ɨɨ**, **ɨ** is regarded as a vowel, and it changes to **-ɨɨyu** according to (2).

Else, **ɨ** is regarded as a consonant, and it changes to **-ɨɨu** according to (1).

Tenses and affirmation/negation of verbs are as below:

	Verb stem ending up with V	Verb stem ending up with C
present negative	*STEM*-**n**	*STEM*-**an**
past affirmative	*STEM*-**taɨ**	*STEM*-**itaɨ**
past negative	*STEM*-**ddan**	*STEM*-**addan**

Answer

(a)

11. *Wallets did not dance.*
12. *Octopi do not make desks.*
13. *Letters flew.*

(7 pts) [1 pt for each argument.]

(b)

14. **innu dusɨɨu miiddan**
15. **mmnu yaayu pusan**
16. **usɨnu pavvu pusaddan**

(13 pts) [1 pt is deducted for each incorrect suffix and 0.5 pts are deducted for each incorrect stem. 0.5 pts are deducted for incorrect word order.]

Reference

木部暢子、トマ・ペラール、林由華、五十嵐陽介、かりまたしげひさ、松浦年男、中島由美、徳永晶子、諸岡大悟（2012）『消滅危機方言の調査・保存のための総合的研究 南琉球宮古方言調査報告書』東京：国立国語研究所。

衣畑智秀、林由華（2014）「琉球語宮古狩俣方言の音韻と文法」,『琉球の方言』, 第 38 巻, 17-49 頁。

Problem 5 Shatranj / Algebraic notation

The basic structure of a move is [piece abbreviation] + [square of arrival]. The square of arrival consists of ranks (**a** to **h**) and files (**1** to **8**).

When a piece makes a capture, a **×** is inserted before the square of arrival.

If a *Soldier* promotes to a *Counselor*, **V** for *Counselor* is indicated after the square of arrival.

If a player calls **shah**, a **+** is added to the end, and if the game ends up with **shah mat**, a **#** is added to the end.

When a game ends, **1-0** is marked for red's win, **0-1** for black's win, and **½–½** or **(=)** for a draw (absent in this problem).

The pieces' name in Romanized modern Persian and their abbreviations are as below:

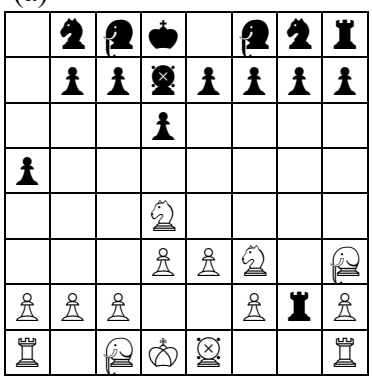
Piece	*King*	*Counselor*	*Elephant*	*Horse*	*Rook*	*Soldier*
Modern Persian	**shah**	**vazir**	**fil**	**asb**	**rakh**	**peyadeh**
Abbreviation	**Š**	**V**	**F**	**A**	**R**	**Ø**

The shaded case is absent in this problem.

If multiple identical pieces can move to the same square, the moving piece is uniquely identified by specifying the file (preferred) or the rank of departure.

Answer

(a)



(6 pts) [1 pt is deducted for each incorrect red horse and 0.5 pts are deducted for each incorrect pieces other than the red horses. 3 points are deducted for a symmetrical reversal (face flip, mirroring of red and black). 2 points are deducted if it is not able to distinguish between red and black.]

(b)

Ae6	**<u>Rad2+</u>**
Šc1	**<u>b2+</u>**
Šb1	**Rd1+**
<u>Ša2</u>	**b1V+**

<u>Šb3</u> **Rd3#**

0-1

(7 pts) [0.5 pts are deducted for each mistake and 2 pts are deducted for each unanswered blank.]

(c)

Rh8+ **Š×h8**

Ff5+ **Šg8**

Rh8+ **Š×h8**

g7+ **Šg8**

Ah6#

1-0

(7 pts) [0.5 pts are deducted for each mistake and 1 pt is deducted for each missing or extraneous move.]

Supplementary Information

The game in (b) is adapted from endgame No. 424 in *A History of Chess*, which was originally the 17th game of the Shatranj manuscript held by the Royal Asiatic Society of Great Britain and Ireland (RAS). The author is *Farazdaq Ỵūnānī*.

The game in (c) is quoted from endgame No. 83 in *A History of Chess*. It was originally recorded in several Shatranj manuscripts and is also known as the "Dilaram Problem" or "The Maiden's Problem". The author is *Abū Bakr al-Ṣūlī*. The following is a translation of the description of subsequent developments in the question:

The red sacrificed his Rook by moving it to **h8**. *The black King is forced to take it. Then the red Elephant moved to* **f5**, *forming a "discovered check". The black King go back to* **g8**. *Now, here comes the brilliant move—the red did that again,* **Rh8+**, *and the black is forced to take it again. After Soldier to* **g7** *check, the black had no choice but move his King one step to the left. Finally, the red Horse stepped forward. Checkmate, or the so-called "***shah mat***". Red won.*

Reference

Murray, H. J. Ruthven. (1913). *A History of Chess.* London: Oxford University Press.